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Addressed to: Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO), National Science Foundation

[Data Science 4 Everyone](#) is a coalition advancing data science and AI education so that every K-12 student is equipped with the data basics needed to compete and succeed in our modern world. Access to data science education is an opportunity to open doors to higher education, high-paying careers, and an engaged community.

Created by the [University of Chicago Center for RISC](#), associated with Freakonomics' Steven Levitt, and organized in partnership with [The Learning Agency](#) and the [Concord Consortium](#), we support a growing community that knows that the data revolution has transformed modern life and we need to prepare our students for it.

As artificial intelligence (AI) permeates our lives and continues to evolve, the significance of data literacy and data science knowledge is tangible. Just as traditional literacy and numeracy are vital, data literacy is now an essential skill for individuals to be able to capitalize on AI benefits in their personal and professional life, understand how to navigate risks, and take advantage of the exploding demand in AI and data-related jobs.

The transformative potential of AI is monumental—it stands to reshape societies, job markets, social interactions, and even political systems. Greater data literacy among all students and more data-savvy graduates to meet workforce demand will better position the US to meet challenges, whether they relate to AI, national security, health, the economy, education, or manufacturing.

If AI is positioned to dramatically transform our economy, from the production line to the c-suite, then everyone must be prepared to leverage its power. AI alone may add between an estimated [\\$2.6 trillion to \\$4.4 trillion annually](#) to the global economy and may automate between 60% to 70% of task-time within existing jobs, rather than full replacement. Earlier studies estimated that emerging technologies will [increase the technology intensity](#) of existing careers across all sectors. A [report by the Burning Glass Institute](#) found that 22% of all jobs in the U.S. economy as of 2023 include at least one “data science skill,” with the highest share of data-skill job postings in utilities (36%), manufacturing (32%), and agriculture (30%). Not every worker will build the next AI algorithm or become a data scientist, but nearly every American will need to leverage

data and AI to maintain a competitive edge in their sector or risk losing entire industries to other countries who do the same. This unprecedented economic growth will only be captured by the countries whose workers are prepared in data and AI basics.

U.S. educators are mostly unsupported to teach students about AI and other emerging technologies. An analysis of math educators nationally found that teachers are least confident to teach about [data and statistics](#), as well as [technology integration](#), compared to other content categories. Computer science was the [least popular credential](#) for K-12 educators to pursue as recently as the 2018–2019 school year. These challenges translate to student opportunities and outcomes. As of 2023, only 5.8% of our high school students [are enrolled](#) in foundational computer science courses. Introductory basics in data or AI are [typically not covered](#) even if they exist in some state standards. As a result, students' foundational data literacy has declined [between one and three grade levels](#) steadily over the past decade, varying disproportionately by race and geography, with losses only accelerated by the pandemic.

China, India, Germany, Canada, and the U.K. have [all made investments](#) to dramatically reshape the student experience for the world of AI and train teachers to educate a modern, digitally-prepared workforce. While the U.S. made early research & development investments in computer science and data science education through the National Science Foundation, we have no teacher workforce ready to implement these innovations in curriculum or educational technology. The number of individuals [completing a teacher preparation](#) program has fallen 25% over the past decade; long-term forecasts suggest [at least 100,000 shortages annually](#), teachers themselves are [discouraging others from joining their own profession](#) (especially in STEM), and preparing to teach digital skills such as computer science was the [least popular option](#) for prospective educators to pursue. There is still consistent evidence that young people or even current professionals remain interested in teaching as a possible career, but only if we create the conditions to translate that interest into action. U.S. policymakers have a narrow window to leverage the strong interest in AI to energize the education workforce, and ensure future graduates are globally competitive for the digital frontier.

To address the critical needs of recruiting and supporting educators to teach AI and data literacy, the Trump Administration should implement the following policy actions:

Recommendation 1. Launch a Grand Challenge for AI-Era Teacher Preparation

Create a national challenge via www.Challenge.Gov to identify the most innovative teacher recruitment, preparation, and training programs to prepare and retain educators for teaching in the era of AI. Challenge requirements should be minimal and flexible to

encourage innovation, but may include the creation of teacher leadership opportunities, peer-network sites for professionals, and digital classroom resource exchanges.

Recommendation 2. Update Areas of National Need

To enable existing scholarship programs to support AI readiness, the U.S. Department of Education should add “Artificial Intelligence,” “Data Science,” and “Machine Learning” to [GAANN Areas of National Need](#) under the Computer Science and Mathematics categories to expand eligibility for Masters-level scholarships for teachers to pursue additional study in these critical areas. The number of higher education programs in Data Science education [has significantly increased](#) in the past five years, with a small but increasing number of emerging Artificial Intelligence programs.

Recommendation 3. Expand and Simplify Key Programs for Technology-Focused Training

The President should direct the U.S. Secretary of Education, the National Science Foundation Director, and the Department of Defense Education Activity Director to add “Artificial Intelligence, Data Science, Computer Science” as competitive priorities where appropriate for existing grant or support programs that directly influence the national direction of teacher training and preparation, including the [STEM Corps](#) (NSF), the [Robert Noyce Scholarship Program](#) (NSF), and the DoDEA [Professional Learning Division](#), and the [Apprenticeship Building America](#) grants from the U.S. Department of Labor. These terms could be added under prior “STEM” competitive priorities, such as the [STEM Education Acts of 2014](#) and 2015 for “Computer Science,” and framed under “Digital Frontier Technologies.”

In addition, the 119th Congress should authorize the National Science Foundation to create two programs to support pre-service and in-service educators:

1. **A national Digital Frontier Teaching Corps**, a three-year fellowship opportunity covering summertime pay for high-potential educators to conduct intensive study in AI, data science, and computing foundations. The DFT Corps would be a prestigious and materially meaningful program to both impart digital technical skills and transform the social perception of the teaching profession. The DFT Corps would include:
 - a. Educator scholarships for full summertime pay
 - b. A low-barrier digital application process for individuals to apply for funding directly from the NSF or an intermediary third-party organization
 - c. Automatic enrollment in local site-based training sites, including college-level coursework (computer science, data science, artificial intelligence, quantum computing) and peer-based pedagogical training

- d. Tax-based verification for program completion status
 - e. Federal loan forgiveness upon successful program completion
 - f. Federal recognition awards of schools employing DFT Corps educators
 - g. Eligibility to apply for DFT Corps training site leadership upon successful program completion
2. **Teacher College Innovation Grants**, a competitive NSF grant program for modernizing teacher preparation programs and teacher licensure models. Teacher College Innovation Grants would provide research funding and capacity to evaluate DFT Corps training sites and ensure lessons learned are quickly integrated back into teacher preparation programs. Competitive priorities would be made for:
- a. Proposals that develop and/or scale new methods-style courses for teaching digital technology (i.e., computational thinking, data literacy, AI literacy) to students across traditional school subjects
 - b. Proposals that develop new models for teacher preparation, leveraging online learning or other digital training systems that result in lower program costs
 - c. Proposals that focus on building or modernizing teacher preparation capacity in majority-rural communities

Lastly, Congress should **direct the NSF to expand focus on translating evidence on AI teaching** to schools and districts by maintaining support for the NSF [Discovery Research PreK-12 Program Resource Center on Transformative Education Research and Translation \(DRK-12 RC\)](#) program and the expand the NSF [Translation and Diffusion \(TD\)](#) grant program. The DRK-12 RC program could be utilized to work with multi-sector constituencies to accelerate the identification and scaling of evidence-based practices for AI, data science, computer science, and other emerging tech fields.

Thank you for considering these actions at an important time in our nation's history. AI and data literacy are crucial to our nation's competitiveness and economic growth.

Please do not hesitate to reach out to **Zarek Drozda, Executive Director of Data Science 4 Everyone** if you have questions or if there is anything that Data Science 4 Everyone can do to support AI and data literacy efforts.

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